

THESIS INFORMATION

Title: **Research on Opinion Mining Approaches for User Feedback**

Major Code: **Computer Science**

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ABSTRACT

Opinion analysis (also known as sentiment analysis) is a research area that has garnered significant attention within both academic and industry communities, due to its potential to extract meaningful information from user-generated content. This dissertation aims to develop various approaches to address opinion analysis tasks on user feedback comments, with the objective of improving performance on Vietnamese datasets as well as those in other languages. The proposed methods leverage the capabilities of pre-trained language models and explore multiple paradigms, including classification-based approaches, generation-based approaches, and ensemble methods built upon language models. Additionally, the dissertation addresses the challenge of limited labeled data in low-resource scenarios by proposing strategies to mitigate data scarcity. The dissertation makes three key contributions, as outlined below:

- Improving the performance for level-based opinion mining tasks: The dissertation proposes three approaches to improve prediction performance across diverse datasets: (1) Classification-based transfer learning leveraging pre-trained language models; (2) Generation-based transfer learning using generative language models; and (3) Ensemble methods based on language models, which integrate predictions from multiple model variants to enhance overall performance.
- Developing enhanced approaches for regular and comparative opinion mining tasks: The dissertation introduces advanced modeling techniques for opinion mining under both regular and comparative perspectives. For regular opinions, a prompt-based generative language model is presented within a multi-task learning framework to jointly extract a quadruple of opinion elements. For comparative opinions, a two-

stage architecture is proposed: the first stage identifies whether a comment expresses a comparative opinion, and the second stage extracts the quintuple of comparative opinion elements.

- Proposing methods for addressing data scarcity in opinion mining for low-resource languages or domains: The dissertation introduces two different approaches to tackle the challenge of limited training data in opinion mining tasks within low-resource settings. (1) Leveraging multilingual pre-trained language models in combination with high-resource language data across various transfer learning scenarios to boost performance in low-resource languages or domains; (2) Designing instruction-based prompts tailored for modern large language models (LLMs) to enable effective learning under limited supervision.

In addition, the PhD candidate also presented the results of a semantic survey study reviewing previous methods, approaches, and datasets related to the opinion mining task to identify limitations that need to be addressed within the dissertation's research scope.

Regarding scientific publications, the PhD candidate has published 9 scientific papers, including 07 papers in reputable journals (comprising 03 articles in SCIE Q2-ranked journals, 02 in international Scopus-indexed journals, and 02 in domestic journals) and 02 papers presented at C-ranked international conferences during the dissertation period.

Supervisor



Nguyễn Lưu Thùy Ngân

PhD Student



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